The kicking function in a Mathematica notebook and in the ChamPBH pipeline: an audit, and the effect of the difference on capture near $\beta = 3$

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16 October 2026

Abstract

A conformally coupled scalar field in the radiation era is pushed, or "kicked", whenever the radiation is not exactly conformal, that is whenever its equation of state departs from $w = 1/3$. Two independent codes integrate this system: a Mathematica notebook (`PeakEWsurf.nb`) and a Python pipeline (ChamPBH). We compare how each one computes the kick. The kick term is algebraically identical in the two codes. Both build the kicking function $\Sigma(T_J) = 1-3w(T_J)$ by interpolating a table of w, but *the tables are not the same above* 22 GeV. Below that temperature every shared row agrees to 10^{-12}. Between 22 and 100 GeV the notebook's Σ is larger by up to 10%. Above 100 GeV the notebook keeps only four rows and forces $w = 1/3$ at 386.7 GeV, so its Σ is exactly zero above that temperature; the pipeline's table continues smoothly to 25 TeV. The two codes also interpolate differently, but above 20 GeV that contributes at most 6×10^{-5} to Σ, against 4×10^{-3} from the rows.

To measure the dynamical effect we re-ran the notebook's own integration with the pipeline's Σ substituted. This moves the coupling above which the field is captured at the electroweak scale from $\beta = 3.00285$ to $\beta = 3.01949$. For $3.003 \lesssim \beta \lesssim 3.019$ the histories change qualitatively. With the notebook's table they are captured at the electroweak peak, near $T_J \approx 52$ GeV; with the pipeline's they pass through it and are captured at the QCD transition, near $T_J \approx 1.64$ GeV. More than 99% of the shift comes from the table rows, and none of it comes from the interpolation. With the pipeline's Σ the notebook reproduces the pipeline's own capture edge, $\beta \in (3.019, 3.020]$. Neither table records how it was made. Separately, the notebook's closed-form matter fraction $f_m(T_J)$ appears to have its $(1+w)$ dependence inverted. This changes f_m by up to 17%, but f_m is at most 4×10^{-4} of Σ at the relevant temperatures, so the error has no effect on the dynamics.

Addendum, 12 November 2026 (§7). The pipeline's table is a free gas of Standard Model particles at their vacuum masses above 100 GeV; the notebook's departs from it as Laine & Meyer's equation of state does. Judgement 1 of §5 is withdrawn. Twenty-nine points supplied on 12 November sample the notebook's curve every 10 GeV from 100 to 380 GeV; the table they make with the notebook's rows is compared with Laine & Meyer's data.

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1 The system and the kick

1.1 Equations of motion

Both codes describe a scalar field ϕ in the Einstein frame, coupled to matter and radiation through the Jordan-frame metric $\tilde{g}_{\mu\nu} = A^2(\phi)\, g_{\mu\nu}$, with

$$A(\phi) = \exp(\beta\phi/M_{\rm P}), \tag{1}$$

where $M_{\rm P} = 2.43 \times 10^{18}\,\text{GeV}$ is the reduced Planck mass. Below we set $M_{\rm P} = 1$. The potential is $V(\phi) = v_0^4 \exp(M/\phi)$. Matter and radiation are minimally coupled to $\tilde{g}$, so their Jordan-frame temperature T_J obeys the usual thermodynamics. The radiation density is $\tilde{\rho}_r = (\pi^2/30)\, g_\rho(T_J)\, T_J^4$, its Einstein-frame density is $\rho_r = A^4\tilde{\rho}_r$, and $w(T_J) = \tilde{p}_r/\tilde{\rho}_r$ is its equation of state. Let $f_m = \rho_m/\rho_r$ be the matter fraction and $y = \mathrm{d}\phi/\mathrm{d}N$, where N counts Einstein-frame e-folds. The Friedmann equation is

$$3H^2 = \frac{\rho_r(1+f_m) + V}{1 - y^2/6}. \tag{2}$$

The field is sourced by the trace of the matter stress tensor:

$$\ddot{\phi} + 3H\dot{\phi} + V'(\phi) = -\frac{\mathrm{d}\ln A}{\mathrm{d}\phi}\, T^{\mu}{}_{\mu} = -\beta\,(\rho_r - 3p_r + \rho_m) = -\beta\rho_r\,(\Sigma + f_m)\,, \tag{3}$$

where

$$\boxed{\Sigma(T_J) \equiv 1 - 3w(T_J)} \tag{4}$$

is the *kicking function*. It vanishes for conformal radiation and becomes positive whenever a species becomes non-relativistic or the plasma passes through a transition. In the Standard Model it has three peaks, at $e^{\pm}$ annihilation ($T_J \approx 0.16\,\text{MeV}$), at the QCD crossover ($\approx 0.18\,\text{GeV}$) and at the electroweak scale ($\approx 53\,\text{GeV}$) (Fig. 1a). Written in e-folds, the kick is the term

$$\left.\frac{\mathrm{d}y}{\mathrm{d}N}\right|_{\rm kick} = -\frac{\beta\rho_r(\Sigma + f_m)}{H^2} = -3\left(1 - \frac{y^2}{6}\right)\frac{\beta\rho_r\,(\Sigma + f_m)}{\rho_r(1+f_m) + V}. \tag{5}$$

The Jordan-frame temperature follows from entropy conservation in the Jordan frame,

$$\frac{\mathrm{d}\ln T_J}{\mathrm{d}N} = -\frac{1+\beta y}{1+\frac{1}{3}\,\mathrm{d}\ln g_s/\mathrm{d}\ln T_J}. \tag{6}$$

So T_J *stops falling and turns round* wherever $1+\beta y = 0$. If the kick drives y below $-1/\beta$ near a peak of Σ, then T_J reaches a minimum and stays near that peak. In this report a history is *captured* at a peak if T_J has a minimum there. In particular, a history is captured at the electroweak scale if its first T_J minimum lies above 20 GeV. This is an operational criterion, chosen for this comparison; it is not the pipeline's own classification. For β close to 3 the outcome is delicate, because y then hovers near $-1/\beta$ for many e-folds.

1.2 How the two codes write the kick

Notebook. The notebook integrates $(\phi, y, \ln T_J)$ with Mathematica's `NDSolve`. The right-hand side for y contains the term

```
-3 (1 - y^2/6) (Vp + beta rho (Sigma[T] + fmTinterp[T]))
                  / (rho (1 + fmTinterp[T]) + V)
```

which is Eq. (5) together with the potential gradient. Here `rho` is computed from T_J at each step as $A^4(\pi^2/30)g_\rho T_J^4$. The notebook defines `Sigma[T] = 1 - 3 w[T]`, with w taken from a table (§2), and f_m from a closed-form expression (§3).

Pipeline. The pipeline integrates $(\phi, y, \ln\rho_r, \ln f_m, \ln T_J)$. Its kick term is

$$\left.\frac{\mathrm{d}y}{\mathrm{d}N}\right|_{\text{kick}} = -3E\,\beta\,\frac{\Sigma + f_m}{1+f_m}, \qquad E = 1 - \frac{y^2}{6} - \frac{V}{3H^2}, \tag{7}$$

with $\Sigma = 1 - 3w(T_J)$ evaluated from its own table. By Eq. (2), $E = \rho_r(1+f_m)/3H^2$. Mathematica's `Simplify` reduces the difference between this and Eq. (5) to exactly zero.

Conclusion 1. *The kick has the same functional form in both codes.* Any difference must come from the inputs: $\Sigma(T_J)$ and $f_m(T_J)$.

2 The tables behind $\Sigma(T_J)$

2.1 What each code tabulates

	Notebook	Pipeline
Rows	132	189
Range of rows	3.65×10^{-5} – 386.7 GeV	10^{-5} – 2.51×10^{4} GeV
Spacing	0.05 dex to 100 GeV, then rows at 158, 251, 316 and 386.7 GeV	0.05 dex throughout
End rows	$w = 1/3$ exactly at both ends, off the 0.05-dex grid	on the grid
Outside the rows	$w = 1/3$	$w = 1/3$
Interpolation	Mathematica `Interpolation` (order 3) of $\ln w$ against $\ln T$: piecewise local cubics, continuous	cubic B-spline of w against $\ln T$ (`scipy make_interp_spline`), C^2

Table 1: The two tabulations of $w(T_J)$.

Table 1 summarises the two tables. Of the notebook's 132 rows, 130 lie on the pipeline's 0.05-dex grid. The other two are its end points, where $w = 1/3$ appears to have been imposed

by hand. Of the 130 shared rows, **113 agree to** 10^{-12}: every shared row below 22 GeV. The 17 that differ are listed in Table 2. From 25 to 100 GeV the notebook's Σ is larger than the pipeline's by 1–10%, the gap growing with temperature. Above 100 GeV the notebook has only three interior rows, which disagree with the pipeline's in both directions. Its final row then forces w back to 1/3 at 386.7 GeV. The pipeline's Σ instead decays smoothly, from 3.4×10^{-3} at 398 GeV to 6×10^{-4} at 1 TeV and 10^{-6} at 25 TeV.

T [GeV]	Σ_{nb}	Σ_{pipe}	$\Delta\Sigma$	T [GeV]	Σ_{nb}	Σ_{pipe}	$\Delta\Sigma$
22.39	0.01888	0.01885	+0.00004	63.10	0.03717	0.03638	+0.00079
25.12	0.02253	0.02225	+0.00028	70.79	0.03603	0.03477	+0.00127
28.18	0.02620	0.02567	+0.00053	79.43	0.03438	0.03251	+0.00187
31.62	0.02968	0.02917	+0.00051	89.13	0.03227	0.02987	+0.00240
35.48	0.03274	0.03220	+0.00054	100.00	0.02978	0.02697	+0.00281
39.81	0.03517	0.03472	+0.00044	158.49	0.01721	0.01559	+0.00161
44.67	0.03678	0.03649	+0.00030	251.19	0.00506	0.00768	−0.00263
50.12	0.03761	0.03732	+0.00029	316.23	0.00121	0.00519	−0.00398
56.23	0.03772	0.03733	+0.00039	386.7	0 (imposed)	0.00363*	

Table 2: The shared table rows that differ, with $\Delta\Sigma = \Sigma_{\mathrm{nb}} - \Sigma_{\mathrm{pipe}}$. The notebook's 386.7 GeV row is not on the pipeline's grid. The value marked * is the pipeline's interpolant there.

2.2 Peaks and integrated kick

Table 3 compares the three peaks and the integral $\int \Sigma \, \mathrm{d} \ln T_J$ over each. The integral is a rough measure of the total kick a peak delivers. At $e^{\pm}$ annihilation and the QCD crossover the two codes agree to 0.1% or better. At the electroweak scale the notebook's peak is 0.9% higher and 1.6% hotter. Between 20 and 386.7 GeV its integral is 1.0% larger. Above 386.7 GeV the notebook delivers no kick, where the pipeline delivers 0.0019. Over 20 GeV–20 TeV as a whole, the notebook's integral is therefore 1.8% *smaller* (0.0673 against 0.0685).

Peak	$\Sigma_{\max}$ (at T_J)		$\int \Sigma \, \mathrm{d} \ln T_J$	
	Notebook	Pipeline	Notebook	Pipeline
$e^{\pm}$ (10^{-5}–2×10^{-3} GeV)	0.10074 (0.1607 MeV)	0.10073 (0.1603 MeV)	0.16087	0.16099
QCD (2×10^{-3}–1 GeV)	0.31448 (0.1820 GeV)	0.31453 (0.1820 GeV)	0.50823	0.50821
EW (10–386.7 GeV)	0.03776 (54.08 GeV)	0.03744 (53.21 GeV)	0.07346	0.07277
Above 386.7 GeV (to 2×10^4 GeV)	0	0.00362 (387 GeV)	0	0.00192

Table 3: Peaks of Σ and its integral over $\ln T_J$, evaluated on a grid of 1000 points per decade.

2.3 Is it the data or the interpolation?

The two codes differ in both their rows and their interpolation scheme. We separated the two effects by interpolating each table with the *other* code's scheme. This gives four versions of Σ: two sets of rows, each with two schemes. Figure 2 shows the largest $|\Delta\Sigma|$ in each 0.05-dex interval when one ingredient is changed and the other is held fixed:

- **Changing the rows** (black) moves Σ by up to 4.1×10^{-3} (at 337 GeV) between 20 and 386.7 GeV, and by up to 3.6×10^{-3} above 386.7 GeV. The result is the same whichever scheme is held fixed. Below 10 GeV the only change is near 40 keV, where the notebook's imposed lower end point replaces the pipeline's rows (up to 5×10^{-4}, well below the $e^{\pm}$ peak).

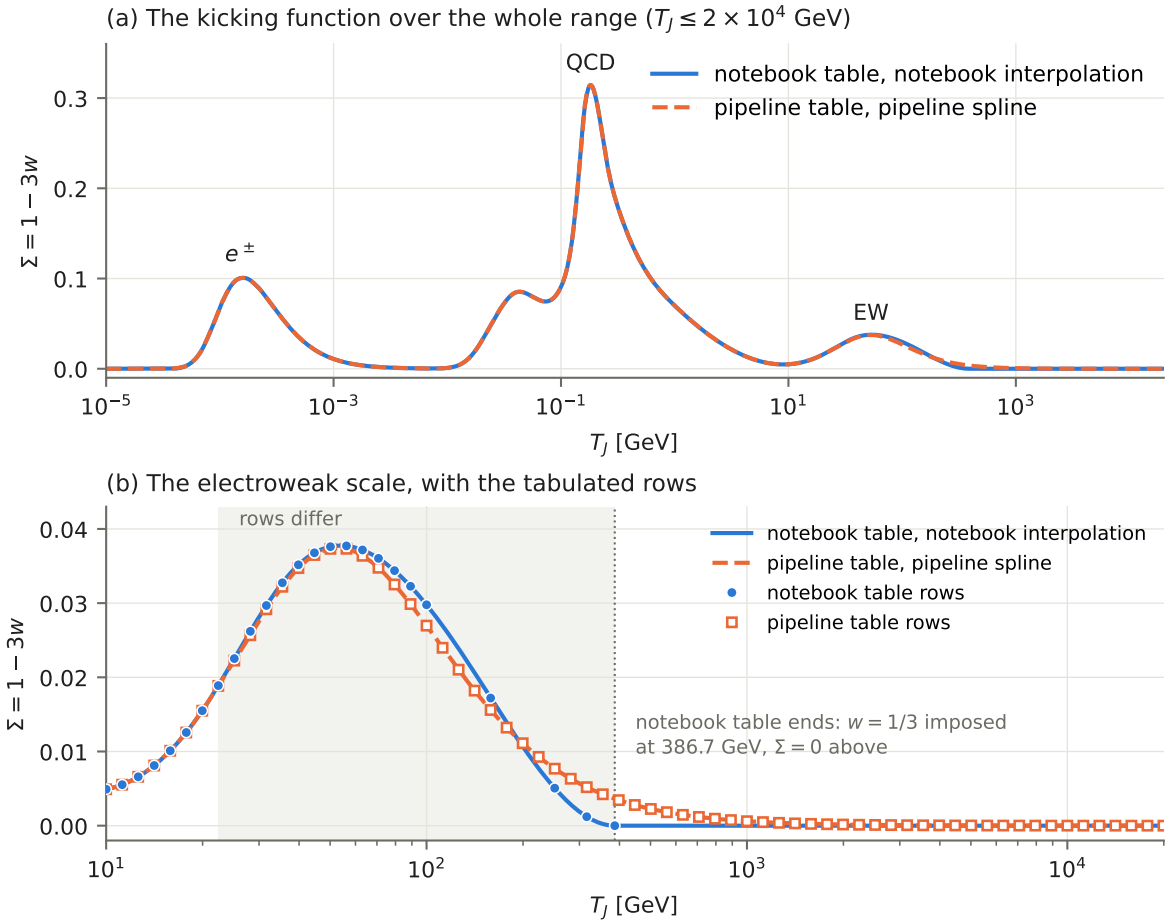


Figure 1: The kicking function $\Sigma = 1 - 3w$ in the two codes. (a) Over the whole range the two curves are indistinguishable at this scale. (b) At the electroweak scale they are not. The markers are the rows each code tabulates. The shaded band marks where shared rows differ, and the dotted line marks the notebook's last row, where $w = 1/3$ is imposed.

- **Changing the interpolation** moves Σ by at most 1.1×10^{-5} on the pipeline's rows and 5.5×10^{-5} on the notebook's sparser rows, anywhere above 20 GeV. The interpolants agree at every row and differ only between rows. The largest scheme difference anywhere is 1.3×10^{-3}, at 0.167 GeV on the steep side of the QCD peak. The QCD integral is unchanged to 2×10^{-5}, and §4.3 shows that this difference has no dynamical effect.

Over 20 GeV–20 TeV the integral $\int \Sigma \, \mathrm{d} \ln T_J$ is 0.06731/0.06729 for the notebook's rows (notebook / pipeline scheme) and 0.06854/0.06854 for the pipeline's.

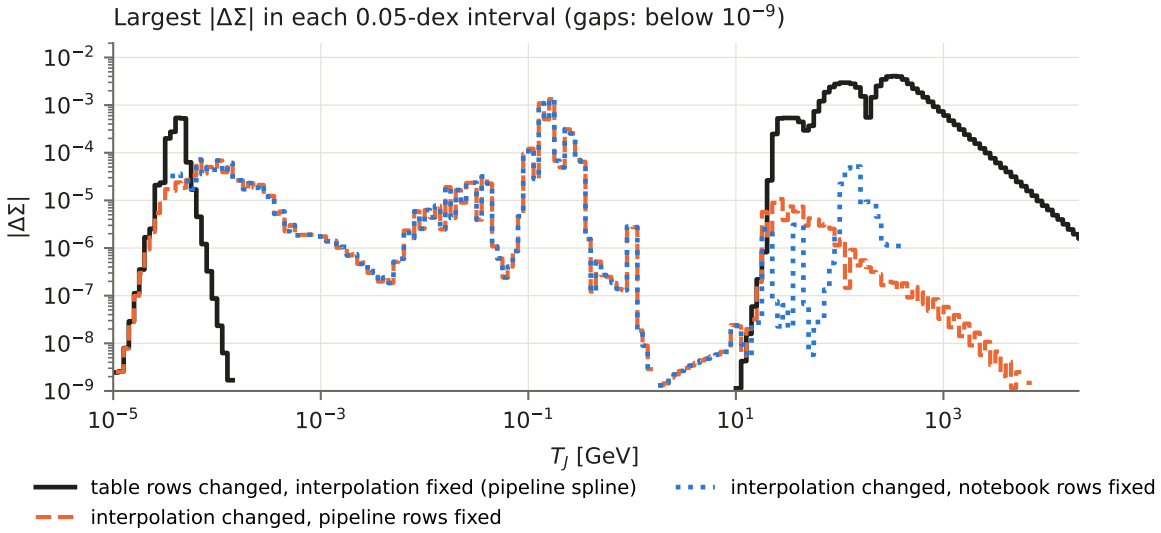


Figure 2: Each curve changes one ingredient of Σ and holds the other fixed. The changed ingredient is either the table rows (black) or the interpolation scheme (dashed and dotted). Above 20 GeV the rows matter 75 to 400 times more than the scheme.

Conclusion 2. *The two codes' $\Sigma(T_J)$ differ at the electroweak scale because their tables differ, not because they interpolate differently.*

3 The matter fraction

The kick is sourced by $\Sigma + f_m$, so we also compared f_m. The notebook uses the closed form

$$f_m^{\mathrm{nb}}(T_J) = \frac{2.78 \times 10^{-9}\,\mathrm{GeV}}{3\,T_J\,\big(1 + w(T_J)\big)}, \tag{8}$$

tabulated with linear interpolation on a 10^{-4}-dex grid. The pipeline instead evolves $\ln f_m$, with $\mathrm{d} \ln f_m / \mathrm{d}N = (1 - \Sigma)(1 + \beta y)$. Its starting value is $f_m = \rho_{m0}(1+z)^3 / \tilde{\rho}_r(T_J)$, where $1 + z$ follows from conservation of $T_J\, g_s^{1/3}$. For the comparison we evaluate the pipeline's f_m from that starting formula at each T_J, i.e. at fixed field.

For a single-temperature plasma, $\tilde{\rho}_r + \tilde{p}_r = T_J s$, which gives

$$f_m = \frac{\rho_m}{s}\,\frac{1+w}{T_J}, \qquad \frac{\rho_m}{s} \simeq 5.2 \times 10^{-10}\,\mathrm{GeV} \text{ (Planck 2018)}. \tag{9}$$

Equation (8) agrees with this at $w = 1/3$, since $2.78 \times 10^{-9}/4 = 6.95 \times 10^{-10} = \frac{4}{3} \times 5.2 \times 10^{-10}$. Its w-dependence, however, is *the inverse* of the one expected: it divides by $(1 + w)$ where the thermodynamic result multiplies by it. Relative to the expected form, the notebook's f_m is too large by $(16/9)/(1+w)^2$, about 1.18 at the QCD peak. Figure 3a shows the ratio of the

two codes' f_m. It ranges from 0.87 (at 10 keV, where the notebook also omits $g_s \neq g_\rho$ after neutrino decoupling) to 1.17 (at 0.19 GeV).

The error does not matter here. Figure 3b shows that f_m/Σ is at most 3.6×10^{-4} between 0.1 MeV and 386 GeV, at the trough near 7 MeV, and is 10^{-8}–10^{-10} at the QCD and electroweak peaks. Above 386.7 GeV, f_m is the notebook's *only* source of kick, but it is between 2×10^{-12} and 3×10^{-14} there. The pipeline's Σ is 3.6×10^{-3} at 387 GeV and still 1.6×10^{-6} at 20 TeV. All the runs in §4 keep the notebook's own f_m.

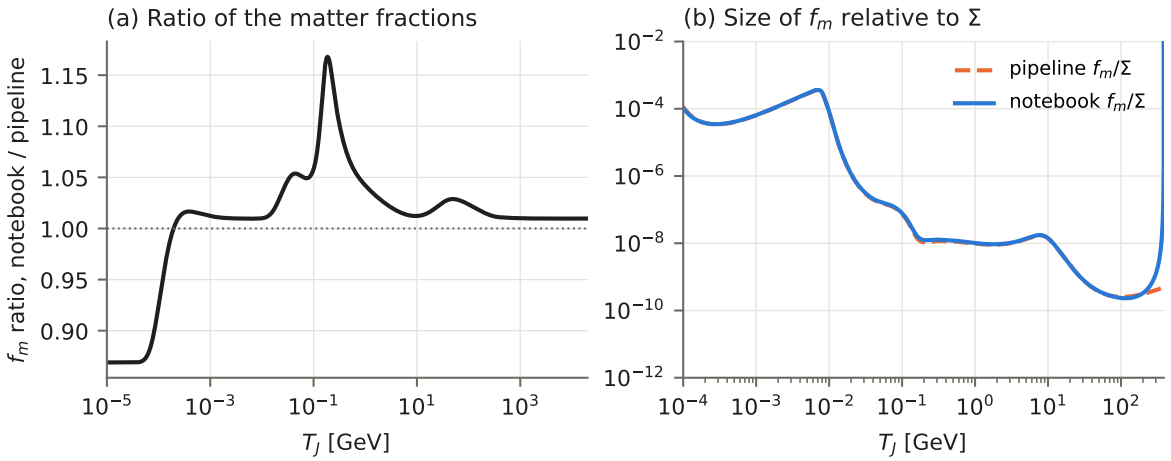


Figure 3: (a) The notebook's matter fraction relative to the pipeline's. (b) The matter fraction as a fraction of Σ. Above 300 GeV the notebook's curve rises because its Σ falls to zero.

4 Dynamical effect near $\beta = 3$

4.1 Method

We re-ran the notebook's own integration, changing nothing but Σ. All other definitions were evaluated directly from the notebook's input cells by a Mathematica 15.0.1 kernel, without transcription. These include the effective degrees of freedom g_ρ and g_s, f_m, V and $\rho_r(T_J)$. The integrator settings are also the notebook's: `NDSolve` with `StiffnessSwitching`, accuracy and precision goals of 10, and `MaxStepSize` 10^{-3}. Every run starts at $T_J = 2 \times 10^4$ GeV with $y = 0$, and stops when ϕ falls to $0.1\,M_P$ or at $N = 150$. The parameters are those of the notebook: $\phi_i = 10\,M_P$ (and $100\,M_P$ for one run), $M = 10^{-2}\,M_P$ and $v_0 = 2.16 \times 10^{-12}$ GeV.

Four versions of Σ were used:

`nb`	the notebook's Σ, unchanged
`pipe`	the pipeline's Σ everywhere
`ew`	the notebook's Σ below 20 GeV, the pipeline's above (the two agree to 3×10^{-6} at 20 GeV)
`csvnb`	the pipeline's rows, interpolated the notebook's way

The pipeline's Σ was evaluated by its own code, sampled at 1000 points per decade, and passed to Mathematica as a cubic interpolant in $\ln T_J$. That interpolant differs from the pipeline's own values by at most 9.4×10^{-9}, measured at the midpoints between samples. Turning points of T_J are the zeros of $1 + \beta y$, located with `FindRoot`.

4.2 Results

Table 4 and Figs. 4–5 give the results. In the window $3.003 \lesssim \beta \lesssim 3.019$ the outcome changes qualitatively. With the notebook's Σ the field is captured at the electroweak peak: T_J reaches

a minimum of 51–53 GeV within 9–12 e-folds and then stays at that scale. With the pipeline's Σ the history passes through the electroweak scale and is captured at the QCD transition, reaching a minimum of 1.64–1.65 GeV 15 to 24 e-folds later. The integration also runs about 3.5 e-folds longer before ϕ reaches $0.1\,M_{\rm P}$.

Outside the window the type of capture is the same, but the details move. Above it, the pipeline's Σ gives a minimum 3.5–4 GeV colder (49–51 GeV against 53–54 GeV), 0.8–2.8 e-folds later. Below it, both are captured at QCD at the same temperature, 1.626–1.631 GeV, but the notebook's history arrives 6–9 e-folds later. At $\beta = 3.000$ the notebook's history has no minimum at all before ϕ reaches $0.1\,M_{\rm P}$; it ends at $T_J = 22.8$ GeV, still falling slowly.

	Notebook Σ			Pipeline Σ		
β	first minimum	N	$N_{\rm end}$	first minimum	N	$N_{\rm end}$
2.990	QCD, 1.626 GeV	29.16	38.93	QCD, 1.626 GeV	22.91	38.93
2.995	QCD, 1.631 GeV	32.87	38.98	QCD, 1.631 GeV	23.81	38.98
3.000	none (ends at 22.8 GeV)	–	36.56	QCD, 1.636 GeV	24.97	39.03
3.005	**EW**, 51.31 GeV	11.30	35.68	**QCD**, 1.641 GeV	26.59	39.07
3.010	**EW**, 52.06 GeV	10.28	35.65	**QCD**, 1.646 GeV	29.09	39.12
3.015	**EW**, 52.68 GeV	9.84	35.67	**QCD**, 1.651 GeV	34.08	39.15
3.020	EW, 53.24 GeV	9.55	35.70	EW, 49.36 GeV	12.37	35.90
3.025	EW, 53.77 GeV	9.34	35.73	EW, 50.15 GeV	10.48	35.82
3.030	EW, 54.26 GeV	9.17	35.76	EW, 50.77 GeV	9.97	35.85

Table 4: The first T_J minimum and the e-fold at which it occurs, for $\phi_i = 10\,M_{\rm P}$. $N_{\rm end}$ is where ϕ reaches $0.1\,M_{\rm P}$. The bold rows change their type of capture.

The threshold. We bisected in β, to a tolerance of 10^{-4}, for the smallest coupling captured at the electroweak scale:

Rows	Interpolation	Electroweak-capture threshold
notebook	notebook	$\beta_c \in (3.002813,\ 3.002891)$
pipeline	notebook	$\beta_c \in (3.019375,\ 3.019453)$
pipeline	pipeline	$\beta_c \in (3.019453,\ 3.019531)$

Substituting the pipeline's Σ raises the threshold by $\Delta\beta_c = +0.0167$. Changing the interpolation with the rows held fixed moves it by less than 1.6×10^{-4}, under 1% of the total. Just below each threshold there is a band of β in which no T_J minimum occurs before the integration stops (the open markers in Fig. 5). With the notebook's Σ the band includes $\beta = 3.000$–3.0028; with the pipeline's it includes $\beta = 3.0175$–3.0194. We did not map its lower edge.

The pipeline's own fine scan of capture near $\beta = 3$ places its electroweak-capture edge in $(3.019, 3.020]$. That scan used a different potential ($\phi_i = 5\,M_{\rm P}$, $M = 10^{-3}\,M_{\rm P}$, $\Lambda = 1$ meV) and the pipeline's own integrator. The notebook run with the pipeline's Σ falls inside that bracket. This suggests that the threshold is fixed mainly by Σ, and that the two integrators otherwise agree here.

A longer history. The notebook also integrates $\phi_i = 100\,M_{\rm P}$ at $\beta = 3.005$ out to $N = 150$ (Fig. 6). With either Σ, T_J first turns at the electroweak scale, with a minimum at $N \approx 9$–10, and reaches a maximum at $N \approx 83$. With the notebook's Σ the history then stays near the electroweak scale, ending at 58 GeV. With the pipeline's Σ, T_J falls away after the maximum, and the history is captured at QCD (1.64 GeV at $N = 134$).

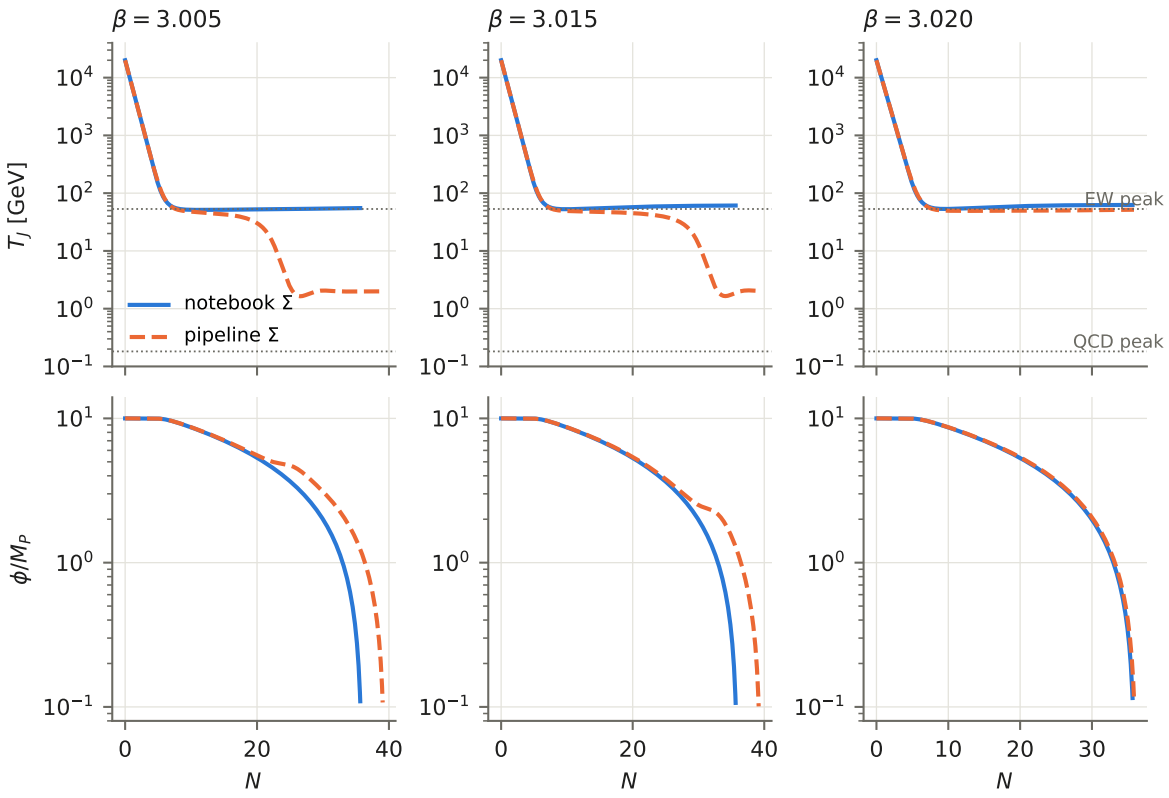


Figure 4: $T_J(N)$ (top) and $\phi(N)$ (bottom) for three couplings, from the notebook's integration with its own Σ (solid) and with the pipeline's (dashed). At $\beta = 3.005$ and 3.015 the notebook's history is captured at the electroweak scale and the pipeline-Σ history at QCD. At $\beta = 3.020$ both are captured at the electroweak scale.

4.3 Attribution, controls and convergence

- **The electroweak table is responsible.** The `ew` variant uses the notebook's Σ below 20 GeV, including its interpolation at the QCD and $e^{\pm}$ peaks. Over all nine couplings of the scan it reproduces the `pipe` histories to $|\Delta \ln T_J| \leq 1.5\times10^{-5}$ and $|\Delta\phi| \leq 5\times10^{-6}\, M_{\rm P}$. So the 1.3×10^{-3} scheme difference on the side of the QCD peak has no effect.

- **The rows are responsible, not the scheme.** The `csvnb` threshold lies within 1.6×10^{-4} of the `pipe` threshold, against a total shift of 0.0167.

- **The substitution is faithful.** Replacing the notebook's definition of Σ with the substituted copy (variant `nb`) changes the unmodified notebook's histories by $|\Delta \ln T_J| \leq 6 \times 10^{-6}$ at $\beta = 3.000$ and 6×10^{-8} at $\beta = 3.015$.

- **The integrations are converged.** Reducing `MaxStepSize` from 10^{-3} to 10^{-4} at $\beta = 3.015$ changes $\ln T_J$ by at most 9×10^{-8} (`nb`) and 7×10^{-6} (`ew`). It changes no turning point and no end point.

Conclusion 3. *The table difference above* 22 GeV *moves the electroweak-capture threshold by* $+0.0167$ *in* β*, and changes the type of capture for* $3.003 \lesssim \beta \lesssim 3.019$*. With the pipeline's table, the notebook reproduces the pipeline's own capture edge.*

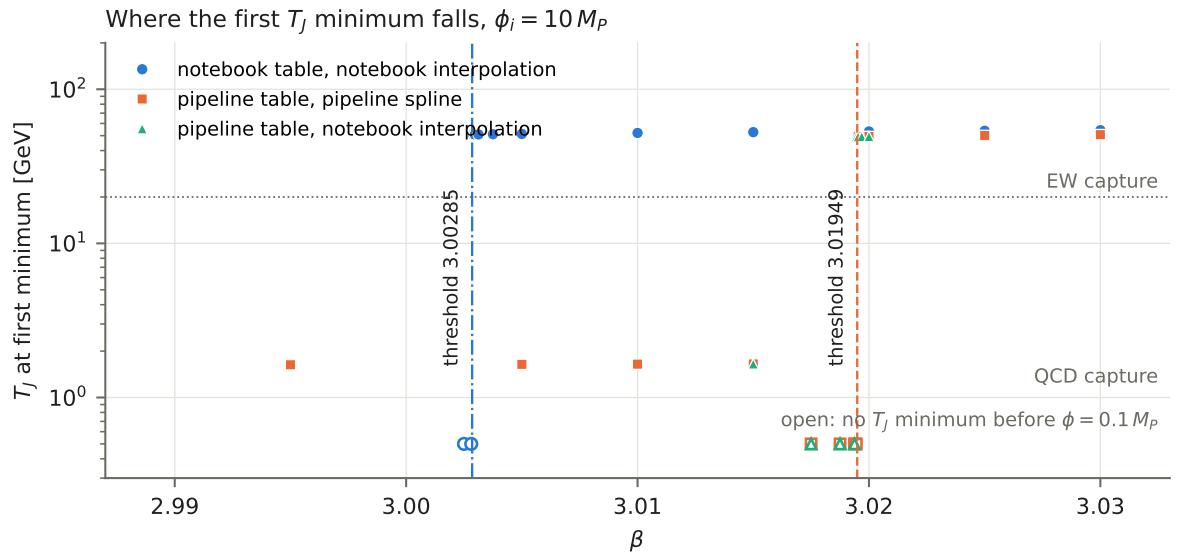


Figure 5: The temperature of the first T_J minimum against β, from the scan and the bisection runs. Filled markers above 20 GeV are electroweak captures and those near 1.6 GeV are QCD captures. Open markers are histories with no minimum before ϕ reaches $0.1\,M_{\mathrm{P}}$. The vertical lines are the two thresholds.

5 Provenance, and which table to trust

Neither code records how its table was made. The pipeline's table was added without the script or method that produced it. It is described as the Saikawa–Shirai equation of state (arXiv:1803.01038) "with adjustments". Its lack of provenance is already an open issue in the pipeline's records. The notebook's table matches it row for row up to 22 GeV, so the two evidently share a source. Above 22 GeV they diverge, and the notebook's rows also disagree with the unadjusted Saikawa–Shirai w. The pipeline keeps that w in a code-comparison table, where it agrees with the pipeline's Saikawa–Shirai implementation to 8×10^{-8} above 10 MeV. Its Σ is 0.0052 at 398 GeV.

Two judgements follow. Neither has been tested beyond what is stated.

1. *The notebook's tail is unlikely to be physical.* At $T \approx 400$ GeV the top quark, Higgs, W and Z are still massive enough to make $\Sigma \sim 10^{-3}$. The running couplings add a trace anomaly that continues above that temperature. A w that reaches $1/3$ exactly at 386.7 GeV and stays there looks like a truncation, not a physical feature. Both the pipeline's adjusted table and the unadjusted Saikawa–Shirai w have a smooth tail there.

2. *We cannot tell which version of the* 25–100 GeV *rows is intended.* The notebook's rows are 1–10% higher there. They may be a different revision of the same adjusted table.

We have not separated how much of the threshold shift comes from the 25–386.7 GeV rows and how much from the missing tail above 386.7 GeV. The two pull in opposite directions in $\int \Sigma \, \mathrm{d} \ln T_J$ (Table 3), and capture is not decided by the integral alone, so the split needs a separate run.

6 Summary and recommendations

1. The kick term is the same in both codes (§1).

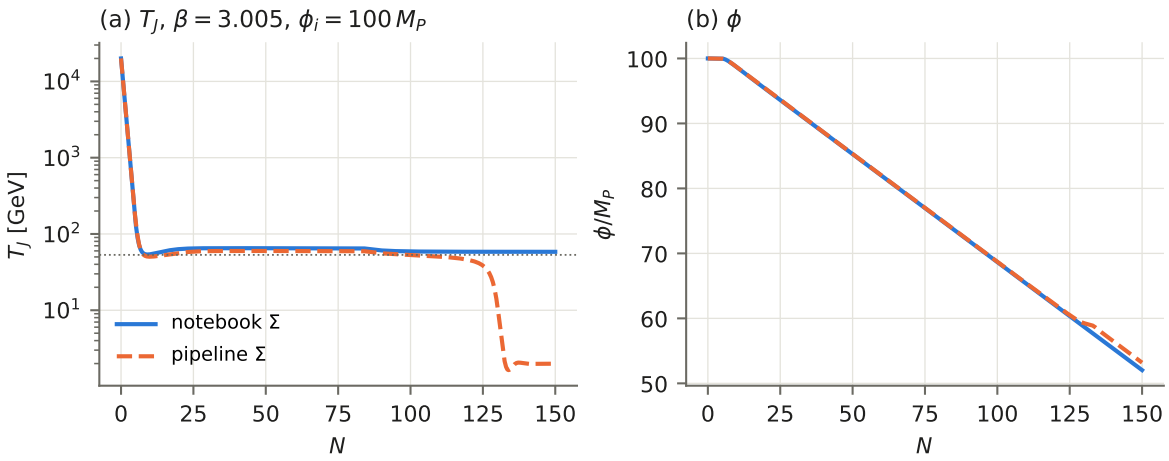


Figure 6: The notebook's $\phi_i = 100\,M_{\mathrm{P}}$, $\beta = 3.005$ history with each Σ.

2. The kicking function differs above 22 GeV, because the tables differ (§2). The interpolation schemes contribute at most 6×10^{-5} there.

3. This moves the electroweak-capture threshold from $\beta = 3.00285$ (notebook) to 3.01949 (pipeline). In between, histories that the notebook captures at the electroweak scale are captured at QCD (§4).

4. The notebook's closed-form f_m appears to have its $(1+w)$ factor inverted (§3). The error is dynamically irrelevant here, but should be fixed before the form is reused.

We recommend:

- Use one $w(T_J)$ table in both codes, with a recorded origin. Until then, results near the electroweak-capture edge should be quoted together with the table that produced them.
- Find out where the notebook's electroweak rows and its 386.7 GeV end point came from. If the truncation was deliberate, record why.
- Run the split described in §5, which needs about three bisections.
- Correct Eq. (8) to $f_m = (\rho_m/s)(1+w)/T_J$, or replace it with the pipeline's treatment, which also includes $g_s \neq g_\rho$.

Not covered. This report compares only Σ, f_m and the form of the kick. It does not compare the two codes' g_ρ and g_s, the T_J equation (6), the friction term, or the treatment of ρ_r (algebraic in T_J in the notebook, integrated in the pipeline). All the runs use the notebook's integrator and parameters; the pipeline's integrator was not run.

7 Addendum (12 November 2026): Laine & Meyer, and a denser notebook table

Everything above was written on 16 October 2026 and stands as written, except where this section says otherwise. Two pieces of information have arrived since.

- On 9 November D. Seery relayed an account from the tables' author. The Saikawa–Shirai fitting functions are meant for g_ρ and g_s alone, not for reconstructing w. The author's

own codes produce w directly, following the references Saikawa and Shirai cite. The notebook's table adds Laine & Meyer's results (arXiv:1503.04935) across the electroweak crossover, whereas the pipeline's table is an ideal gas there.

- On 12 November D. Seery supplied 29 values of w at $T = 100, 110, \ldots, 380\,\mathrm{GeV}$ (`data/notebook_w_points_2026-11-12.csv`, verbatim). D. Seery's understanding, which the author has not confirmed to us, is that they sample a smooth curve drawn to follow Laine & Meyer's data as closely as possible, because those data cannot be interpolated as they stand.

Every number in this section is in `data/laine_meyer_stats.json`, written by `scripts/laine_meyer_compare.py`.

7.1 The pipeline's table is an ideal gas above the crossover

We computed Σ for a free gas of the Standard Model's particles at their vacuum masses, with no interactions and no thermal masses: the photon, the gluons, W, Z, H, the six quarks, the three charged leptons and three neutrinos. On the pipeline's rows from 100 GeV to 25 TeV the pipeline's Σ agrees with it to within 3.3% of Σ; it is 0.0270 against 0.0263 at 100 GeV and 0.00519 against 0.00506 at 316 GeV. Below 60 GeV the pipeline's Σ exceeds the free gas by at most 2.1×10^{-3}. So above the crossover the pipeline's table keeps the heavy particles' vacuum masses. In the symmetric phase they have none, and Σ falls much faster. That is what the notebook's rows do (Fig. 7).

Correction to §5. Judgement 1 ("the notebook's tail is unlikely to be physical") rested on the same vacuum-mass reasoning, and is withdrawn. The notebook's fall to $\Sigma \approx 0$ near 390 GeV is the physical behaviour, to within Laine & Meyer's stated uncertainty (below). Judgement 2 is answered by the author's account: the notebook's 25–100 GeV rows are the newer revision. The recommendation to find out where the rows came from is answered in part; how the curve was drawn is still not recorded.

7.2 Laine & Meyer's data

Laine & Meyer publish their equation of state as one file, `standardmodel.dat` (update of December 2019), at `http://www.laine.itp.unibe.ch/eos15/`. It was downloaded on 9 November 2026 and is kept as `data/laine_meyer_eos15_standardmodel.dat`; its SHA-256 is `457f9beba66f4e4442d012e37eb30506ed018cfada609017e101d691243e64e3`. It has 1720 rows from 10 keV to 10^{18} GeV and gives $w = p/e$ directly (column 6), together with p, e, s, c, c_s^2 and the effective numbers of degrees of freedom.

Above 50 GeV it joins three calculations of different orders, and leaves two gaps between them:

T [GeV]	rows	spacing	source
50–110	18	0.0199 in $\log_{10} T$	hep-ph/0603048
110–125.3	–	no data	
125.3–269.7	434	0.333 GeV	1503.04935
269.7–292	–	no data	
292–1000	355	2 GeV	hep-ph/0510375, modified in 1503.04935

The authors state a theoretical uncertainty of about 1% and advise against taking numerical derivatives, because the pieces do not join smoothly. Since $\Sigma = 1 - 3p/e$, a 1% shift in p/e moves Σ by about 0.01. The joins show it: across $110 \to 125.3$ GeV Σ steps from 0.0274 to 0.0225, and across $269.7 \to 292$ GeV from 0.0041 to 0.0008. There is also a cusp at $T \approx 159$ GeV, where the crossover is. Above 357 GeV Laine & Meyer's Σ is negative, reaching -3.9×10^{-4} at

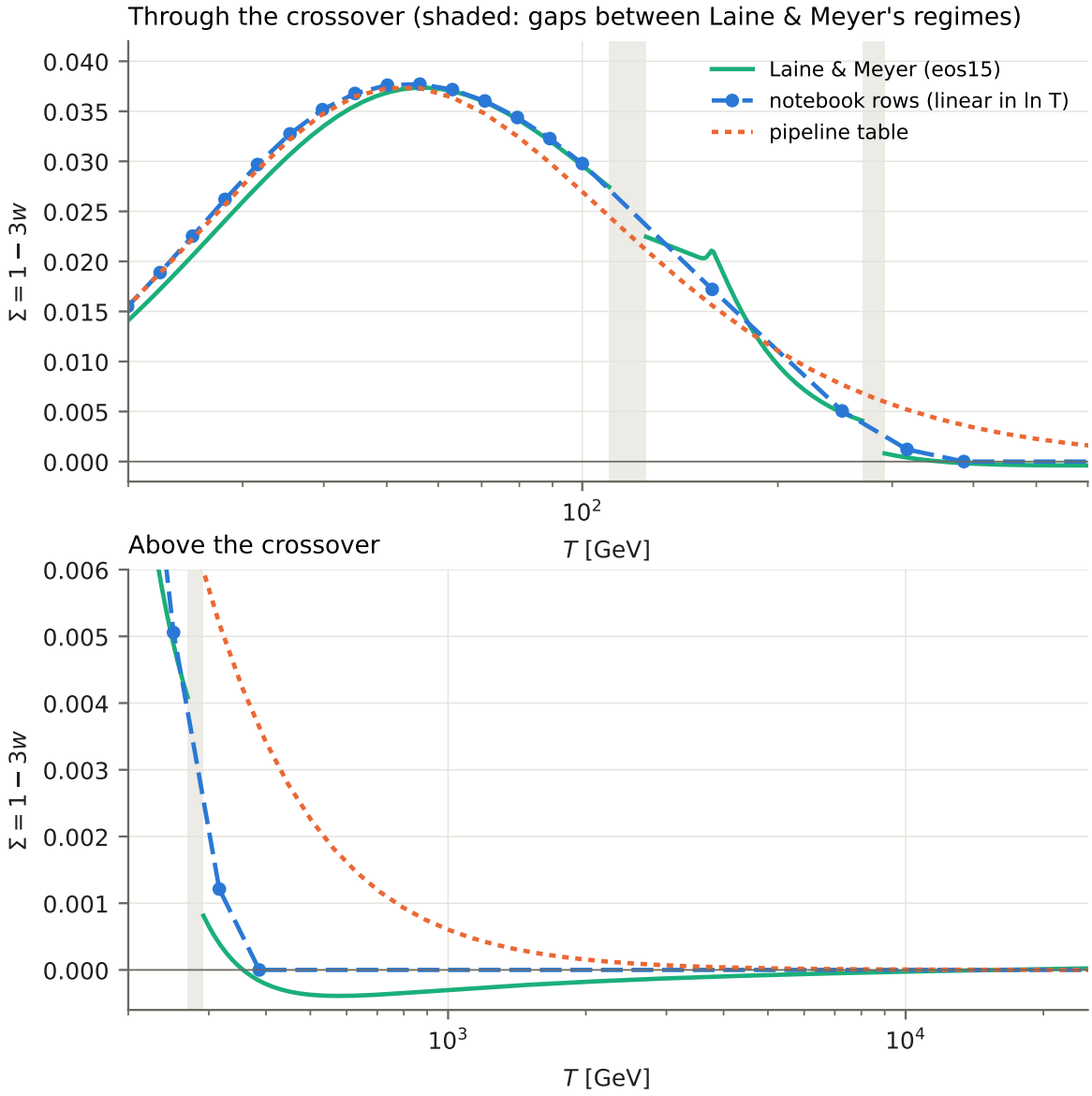


Figure 7: $\Sigma = 1 - 3w$ from Laine & Meyer's data (by regime; shaded, the gaps between them), the notebook's rows (joined linearly in $\ln T$, zero from 386.7 GeV) and the pipeline's table.

574 GeV, until it turns positive again at 15–16 TeV. The paper attributes such features above 300 GeV at least partly to the truncation at $O(g^4)$. Splined as the pipeline splines its table, these data would put spikes into $\mathrm{d}w/\mathrm{d}\ln T$ at the joins and the cusp.

Below 30 GeV the notebook's and the pipeline's rows, which are identical there, differ from Laine & Meyer by 10–30% of Σ through the QCD region. That is a different choice of QCD input, and is outside this addendum's question.

7.3 The points of 12 November, and the merged table

The new points lie on the same curve as the notebook's existing rows. A spline through them reproduces the notebook's rows at 100, 251.2 and 316.2 GeV to 10^{-11} in Σ, and the row at 158.5 GeV to 3.7×10^{-6}, which is the spline's own error between 150 and 160 GeV. They add resolution to the notebook's table; they do not change it.

We call *the merged table* the notebook's 127 rows below 100 GeV, the 29 new points, and

the notebook's row $w = 1/3$ at 386.7 GeV: 157 rows from 3.65×10^{-5} to 386.7 GeV (`data/merged_w_table.csv`). The notebook's rows at 158.5, 251.2 and 316.2 GeV are left out, being on the same curve. Splined as `Xav_EOS_spline` splines its table (a cubic spline in $\ln T$, $w = 1/3$ at and beyond the ends), it behaves well. Above 40 GeV, Σ has its single maximum at 54 GeV and falls steadily to the last row. It is nowhere below -2×10^{-15}, and $\mathrm{d}\Sigma/\mathrm{d}\ln T$ is 2×10^{-8} at the last row, so the switch to $w = 1/3$ leaves no step in the derivative. The derivative is continuous where the spacing changes at 100 GeV. `Xav_EOS_spline` would take the merged table without a change of code.

Against Laine & Meyer (Fig. 8), on their rows:

- 50–110 GeV: higher by 3×10^{-5} to 7×10^{-4}.
- 125–270 GeV: between -3.9×10^{-3} (at the cusp, 159 GeV) and $+1.5 \times 10^{-3}$ (at 125 GeV). Away from 150–170 GeV the difference is within $\pm 1.1 \times 10^{-3}$.
- 292–1000 GeV: higher by 1.5×10^{-3} at 292 GeV, falling to 1.4×10^{-4} at 376 GeV. Above 386.7 GeV the merged table is zero and does not follow the negative tail.

Every difference is smaller than the ~ 0.01 in Σ that Laine & Meyer's stated uncertainty implies, and smaller than the steps at their own joins. The pipeline's table is further away above 250 GeV: at 300–380 GeV its Σ is 3.8–5.7 $\times 10^{-3}$, where Laine & Meyer's is below 7×10^{-4}.

The integrated kick, $\int \Sigma \, \mathrm{d}\ln T$ (the notebook's rows here joined linearly, as in Fig. 7):

T [GeV]	Laine & Meyer	merged	notebook rows	pipeline
30–100	0.04079	0.04183	0.04177	0.04060
100–160	0.01115	0.01110	0.01098	0.00987
160–300	0.00534	0.00542	0.00560	0.00624
300–1000	−0.00028	0.00016	0.00021	0.00279
1000–25000	−0.00031	0	0	0.00031

Over 100–300 GeV the merged table is within 0.2% of Laine & Meyer. Capture is not decided by the integral alone (§5), so the table is a summary, not a prediction of the threshold.

7.4 What this addendum does not settle

- The electroweak-capture threshold with the merged table has not been measured, in either code. The merged table is the notebook's curve, more finely sampled, so the expectation is a threshold close to the notebook's 3.00285; that is not a measurement.
- How the curve between 100 and 386.7 GeV was drawn is not recorded here.
- The pipeline's g_ρ and g_s remain the Saikawa–Shirai fits, which by the author's account are not meant to be consistent with any w; the inconsistency is measured in the companion report on the field equations (`.documents/peakewsurf-field-equations/`).

Note of 13 November 2026. The ChamPBH campaign `electroweak-eos` has since installed the table that follows Laine & Meyer through the crossover as the production w table, with `VERSION_LABEL` 2026.11.0, and has measured the first item above in ChamPBH. On the new table, at $M = 10^{-3}\,M_\mathrm{P}$ and $\phi_* = 5\,M_\mathrm{P}$, with the tuning of `tools/surfing_scan.py`, the electroweak-capture threshold is $\beta \in (3.002,\ 3.003]$, beside the notebook's 3.00285 quoted above. On the table before the campaign, the same tool gives $(3.019,\ 3.020]$. These are ChamPBH's own integrator, not the notebook's. The measurements are in `.documents/electroweak-eos-verification.md`, §2.2.

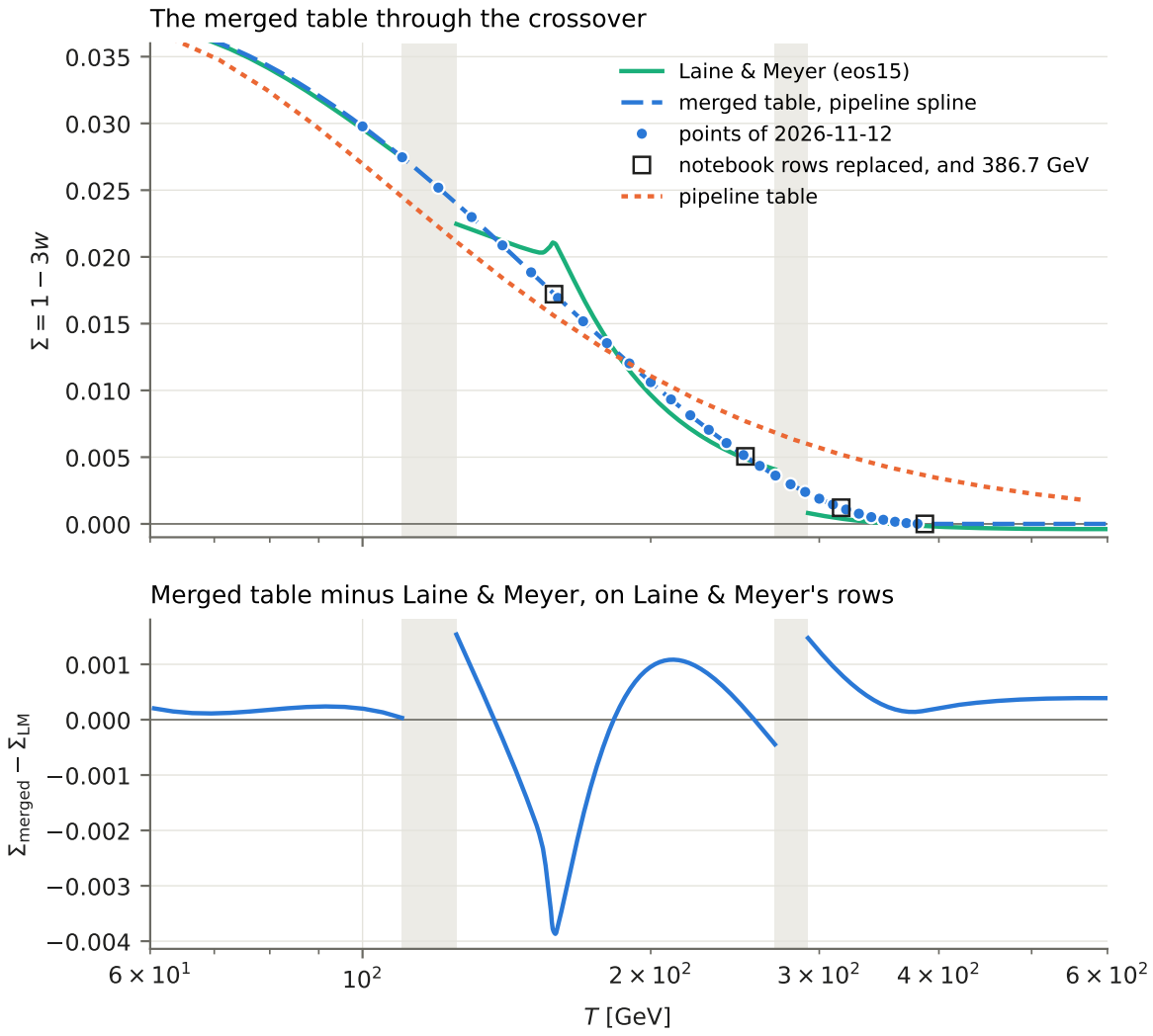


Figure 8: Top: the merged table, splined as the pipeline splines its table, with the points of 12 November, the notebook's rows they replace, Laine & Meyer's data and the pipeline's table. Bottom: the merged table minus Laine & Meyer on Laine & Meyer's rows.

A Reproducing these results

Versions.

- Notebook `PeakEWsurf.nb`, not under version control. Its SHA-256 is `5bd78e6f43b6b7fc4a2eaff746263b63a66c27786eef76cea90edbdd070e92e4`.
- ChamPBH at commit `c21cdfb`.
- Mathematica 15.0.1 (macOS, ARM); Python 3.13.16 with NumPy 2.3.5, SciPy 1.17.0, pandas 3.0.0 and Matplotlib 3.10.8.

Files. Everything is in `.documents/peakewsurf-kicking-function/`:

- `scripts/nb_kick.wls`: evaluates the notebook's definitions and exports its w table, Σ and f_m on a grid, together with the symbolic check of §1.
- `scripts/compare_kick.py`: evaluates the pipeline's Σ and f_m on the same grid. It replaces `datastorekit` with a stub because only the EOS is needed.

- `scripts/sigma_cross.wls`: interpolates the pipeline's rows with the notebook's scheme.
- `scripts/swap_defs.wl`, `swap_runs.wls`, `bisect.wls`: the integrations and the bisection.
- `scripts/compare_runs.py`, `build_data.py`, `make_figures.py`: the analysis, the data products and the figures.
- `data/`: the notebook's table, the Σ/f_m grid at 200 points per decade, the row-by-row comparison, the downsampled histories, `runs_summary.json` (every run's turning points) and `stats.json` (every number quoted here, at full resolution).

Commands. Run from the repository root. `WORK` is a scratch directory, and `NB` is the notebook's absolute path. `wolframscript` must be told where the kernel is.

```
APP="/Applications/Mathematica 15/Wolfram.app"
export WolframKernel="$APP/Contents/MacOS/WolframKernel"
D=.documents/peakewsurf-kicking-function/scripts
wolframscript -file $D/nb_kick.wls $NB $WORK
./venv/bin/python $D/compare_kick.py $WORK    # writes WORK/compare.csv
# then write WORK/sigma_pipe.csv: columns T_GeV, Sigma_pipe of compare.csv
wolframscript -file $D/sigma_cross.wls $NB $WORK $WORK/sigma_pipe.csv \
    "{3}" '{"nb"}' "10^-3" 10 x False
wolframscript -file $D/swap_runs.wls $NB $WORK/runs $WORK/sigma_pipe.csv \
    "{2.99,2.995,3.0,3.005,3.01,3.015,3.02,3.025,3.03}" '{"nb","pipe","ew"}' \
    "10^-3" 10 scan False > $WORK/runs/log.txt
wolframscript -file $D/bisect.wls $NB $WORK/bis $WORK/sigma_pipe.csv \
    "{2.995,3.005}" '{"nb"}' "10^-3" 10 bis False "10^-4" > $WORK/bis/log.txt
```

The remaining runs follow the same pattern: the controls (variant `raw`), the $\phi_i = 100$ run, the step-size check, and the `pipe` and `csvnb` bisections over $(3.015, 3.02)$, the last written to `WORK/bis2`. `build_data.py` expects all the integration sets in one `WORK/runs/log.txt`, each preceded by a header line (`### scan phi_i=10`, `### control raw`, `### phi_i=100`, `### convergence 1e-4`). `build_data.py WORK` reads these logs, and `make_figures.py` draws the figures from `data/`. A single integration takes about 13 s with `MaxStepSize` 10^{-3}. In `bisect.wls` the history files are named to three decimal places, so the bisection steps overwrite one another; the bisection results are taken from its log.

The addendum (§7). ChamPBH at commit `edf72b3`, with the same Python and packages. From the repository root,

```
./venv/bin/python \
    .documents/peakewsurf-kicking-function/scripts/laine_meyer_compare.py
```

takes a few seconds. It reads, in `data/`,

- `laine_meyer_eos15_standardmodel.dat`,
- `notebook_w_table.csv` and `notebook_w_points_2026-11-12.csv`,

and the pipeline's table, and writes

- `data/merged_w_table.csv` and `data/laine_meyer_stats.json`,
- `figures/lm_overview.pdf` and `figures/lm_merged.pdf`.